

Foundations of Image segmentation

[Image segmentation]

1.0 Content Block

Compression-based methods Compression based methods postulate that the optimal segmentation is the one that minimizes, over all possible segmentations, the coding length of the data. The connection between these two concepts is that segmentation tries to find patterns in an image and any regularity in the image can be used to compress it. The method describes each segment by its texture and boundary shape. Each of these components is modeled by a probability distribution function and its coding length is computed as follows:

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Graph partitioning methods Graph partitioning methods are an effective tools for image segmentation since they model the impact of pixel neighborhoods on a given cluster of pixels or pixel, under the assumption of homogeneity in images. In these methods, the image is modeled as a weighted, undirected graph. Usually a pixel or a group of pixels are associated with nodes and edge weights define the (dis)similarity between the neighborhood pixels. The graph (image) is then partitioned according to a criterion designed to model "good" clusters. Each partition of the nodes (pixels) output from these algorithms are considered an object segment in the image; see Segmentation-based object categorization. Some popular algorithms of this category are normalized cuts, random walker, minimum cut, isoperimetric partitioning, minimum spanning tree-based segmentation, and segmentation-based object categorization.

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The functional value is the sum of the total length of the segmentation curve K , the smoothness of the approximation u , and its distance to the original image f . The weight of the smoothness penalty is adjusted by $\mu > 0$. The Potts model is often called piecewise constant Mumford-Shah model as it can be seen as the degenerate case $\mu \rightarrow \infty$. The optimization problems are known to be NP-hard in general but near-minimizing strategies work well in practice. Classical algorithms are graduated non-convexity and Ambrosio-Tortorelli approximation.

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In digital image processing and computer vision, image segmentation is the process of partitioning a digital image into multiple image segments, also known as image regions or image objects (sets of pixels). The goal of segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze. Image segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in images. More precisely, image segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics. The result of image segmentation is a set of segments that collectively cover the entire image, or a set of contours extracted from the image (see edge detection). Each of the pixels in a region are similar with respect to some characteristic or computed property, such as color, intensity, or texture. Adjacent regions are significantly different with respect to the same characteristic(s). When applied to a stack of images, typical in medical imaging, the resulting contours after image segmentation can be used to create 3D reconstructions with the help of geometry reconstruction algorithms like marching cubes.

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Pick K cluster centers, either randomly or based on some heuristic method, for example K-means++
 Assign each pixel in the image to the cluster that minimizes the distance between the pixel and the cluster center
 Re-compute the cluster centers by averaging all of the pixels in the cluster
 Repeat steps 2 and 3 until convergence is attained (i.e. no pixels change clusters)
 In this case, distance is the squared or absolute difference between a pixel and a cluster center. The difference is typically based on pixel color, intensity, texture, and location, or a weighted combination of these factors. K can be selected manually, randomly, or by a heuristic. This algorithm is guaranteed to converge, but it may not return the optimal solution. The quality of the solution depends on the initial set of clusters and the value of K . The Mean Shift algorithm is a technique that is used to partition an image into an unknown apriori number of clusters. This has the advantage of not having to start with an initial guess of such parameter which makes it a better general solution for more diverse cases.

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Disadvantages of MAP and EM based image segmentation
 Exact MAP estimates cannot be easily computed. Approximate MAP estimates are computationally expensive to calculate. Extension to multi-class labeling degrades performance and increases storage required. Reliable estimation of parameters for EM is required for global optima to be achieved. Based on method of optimization, segmentation may cluster to local minima.

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Supervised image segmentation using MRF and MAP
 In terms of image segmentation, the function that MRFs seek to maximize is the probability of identifying a labelling scheme given a particular set of features are detected in the image. This is a restatement of the maximum a posteriori estimation method.

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$$\frac{\partial^2 f(x, y)}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y) \quad \{\displaystyle \frac{\partial^2 f(x,y)}{\partial y^2} = f(x,y+1)+f(x,y-1)-2f(x,y)\}$$

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$$\operatorname{argmin}_{u, K} \gamma |K| + \mu \int_K C | \nabla u |^2 dx + \int (u - f)^2 dx. \quad \{\displaystyle \operatorname{argmin}_{u,K} \gamma |K| + \mu \int_K C | \nabla u |^2 dx + \int (u-f)^2 dx.\}$$

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The Laplacian operator is employed such that the partial derivatives are derived from a specific equation. The second partial derivative of $f(x, y)$ with respect to x and y are given by:

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Image segmentation and primal sketch
 There have been numerous research works in this area, out of which a few have now reached a state where they can be applied either with interactive manual intervention (usually with application to medical imaging) or fully automatically. The following is a brief overview of some of the main research ideas that current approaches are based upon. The nesting structure that Witkin described is, however, specific for one-dimensional signals and does not trivially transfer to higher-dimensional images. Nevertheless, this general idea has inspired several other authors to investigate coarse-to-fine schemes for image segmentation. Koenderink proposed to study how iso-intensity contours evolve over scales and this approach was investigated in more detail by Lifshitz and Pizer. Unfortunately, however, the intensity of image features changes over scales, which implies that it is hard to trace coarse-scale image features to finer scales using iso-intensity information. Lindeberg studied the problem of linking local extrema and saddle points over scales, and proposed an image representation called the scale-space primal sketch which makes explicit the relations between structures at different

scales, and also makes explicit which image features are stable over large ranges of scale including locally appropriate scales for those. Bergholm proposed to detect edges at coarse scales in scale-space and then trace them back to finer scales with manual choice of both the coarse detection scale and the fine localization scale. Gauch and Pizer studied the complementary problem of ridges and valleys at multiple scales and developed a tool for interactive image segmentation based on multi-scale watersheds. The use of multi-scale watershed with application to the gradient map has also been investigated by Olsen and Nielsen and been carried over to clinical use by Dam. Vincken et al. proposed a hyperstack for defining probabilistic relations between image structures at different scales. The use of stable image structures over scales has been furthered by Ahuja and his co-workers into a fully automated system. A fully automatic brain segmentation algorithm based on closely related ideas of multi-scale watersheds has been presented by Undeman and Lindeberg and been extensively tested in brain databases. These ideas for multi-scale image segmentation by linking image structures over scales have also been picked up by Florack and Kuijper. Bijaoui and Rué associate structures detected in scale-space above a minimum noise threshold into an object tree which spans multiple scales and corresponds to a kind of feature in the original signal. Extracted features are accurately reconstructed using an iterative conjugate gradient matrix method.